



Imaging Review of Gastrointestinal Motility Disorders

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Abbreviations: CIPO = chronic intestinal pseudo-obstruction, DES = distal esophageal spasm, EGJO = esophagogastric junction outflow obstruction, LES = lower esophageal sphincter, UES = upper esophageal sphincter

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Review the pathophysiology and clinical symptoms of gastrointestinal motility disorders.
- Describe how motility tests are performed and their results interpreted.
- List the radiologic findings of common motility disorders throughout the gastrointestinal tract.

See rsna.org/learning-center-rg.

The motor function of the gastrointestinal tract relies on the enteric nervous system, which includes neurons spanning from the esophagus to the internal anal sphincter. Disorders of gastrointestinal motility arise as a result of disease within the affected portion of the enteric nervous system and may be caused by a wide array of underlying diseases. The etiology of motility disorders may be primary or due to secondary causes related to infection or inflammation, congenital abnormalities, metabolic disturbances, systemic illness, or medication-related side effects. The symptoms of gastrointestinal dysmotility tend to be nonspecific and may cause diagnostic difficulty. Therefore, evaluation of motility disorders requires a combination of clinical, radiologic, and endoscopic or manometric testing. Radiologic studies including fluoroscopy, CT, MRI, and nuclear scintigraphy allow exclusion of alternative pathologic conditions and serve as adjuncts to endoscopy and manometry to determine the appropriate diagnosis. Additionally, radiologist understanding of clinical evaluation of motility disorders is necessary for guiding referring clinicians and appropriately imaging patients. New developments and advances in imaging techniques have allowed improved assessment and diagnosis of motility disorders, which will continue to improve patient treatment options.

Online supplemental material is available for this article.

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Introduction

The enteric nervous system is critical for gastrointestinal function. The enteric nervous system includes the myenteric (Auerbach) plexus—located between the inner circular muscle layer and the outer longitudinal muscle layer—and the submucosal (Meissner) plexus. The myenteric plexus coordinates the muscular contractions of peristalsis, while the submucosal plexus controls secretion and absorption throughout the gastrointestinal tract (1).

Gastrointestinal motility disorders are common and occur as a result of dysfunction or absence of the myenteric plexus within a segment of the gastrointestinal tract. Dysmotility occurs when the intestinal tract loses the ability to coordinate muscular contractions, thereby leading to clinical symptoms. Motility disorders may be related to primary (idiopathic) or secondary causes, including congenital abnormalities (eg, Hirschsprung disease), inflammatory or degenerative conditions (eg, achalasia), medication-related side effects (eg, opioids), or systemic illness (eg, diabetes mellitus). However, symptoms tend to be nonspecific and may include abdominal

TEACHING POINTS

- High-resolution esophageal manometry is the standard of reference for diagnosing primary esophageal motility disorders and uses closely spaced pressure-sensitive sensors.
- The most frequent symptoms of primary and secondary achalasia include progressive dysphagia to solids and liquids, heartburn, regurgitation, and noncardiac chest pain. The diagnosis should be considered in patients presenting with dysphagia, heartburn that is unresponsive to proton-pump inhibitor therapy, retained food in the esophagus at endoscopy, and resistance to endoscope passage across the gastroesophageal junction. In addition, patients with achalasia are at increased absolute risk for development of esophageal squamous cell carcinoma.
- Gastroparesis is usually associated with variable upper gastrointestinal tract symptoms, including nausea, vomiting, bloating, and early satiety. The diagnosis is made with gastric-emptying scintigraphy, and gastric retention of greater than 60% at 2 hours or greater than 10% at 4 hours is considered abnormal.
- Scleroderma manifests as acute intestinal pseudo-obstruction or CIPO (luminal dilatation can be quite massive), pneumatosis cystoides intestinalis, SIBO, and asymmetric wall fibrosis resulting in wide-mouth jejunal diverticula.
- Dyssynergic defecation results from inappropriate contraction or lack of relaxation of the puborectalis muscle or external anal sphincter, preventing normal passage of stool. A combination of the clinical history and diagnostic tests including anorectal manometry, the rectal balloon expulsion test, electromyography, and defecography is used for diagnosis.

pain, bloating, nausea, heartburn, early satiety, postprandial fullness, or constipation. Therefore, clinical diagnosis of motility disorders frequently involves a combination of the clinical history, radiologic studies, endoscopy, and/or manometry.

This article discusses the common tests used to evaluate and diagnose motility disorders and describes findings of common motility disorders throughout the gastrointestinal tract. Pharyngeal motility disorders have previously been reviewed (2).

Imaging Modalities

Gastric-Emptying Scintigraphy

Gastric-emptying scintigraphy is a functional test to evaluate patients with postprandial nausea and vomiting and early satiety. Patients are required to fast for a minimum of 6 hours, and medications that affect gastric emptying (eg, prokinetics, opiates) should be stopped at least 48 hours before the examination. Hyperglycemia may cause delayed gastric emptying; therefore, fasting glucose levels are checked before performing the examination in patients with diabetes.

Gastric scintigraphy involves ingestion of a low-fat meal, consisting of technetium 99m (^{99m}Tc) sulfur colloid-labeled egg whites, bread, and water within 10 minutes. If the patient is un-

able to consume the meal, the standard reference ranges for gastric emptying will not apply. Imaging is performed with a gamma camera immediately after meal ingestion and again at 1, 2, and 4 hours after ingestion in anterior and posterior projections. Regions of interest are drawn around the stomach; the processing calculation uses a geometric mean, defined as the square root of the product of the anterior and posterior counts (Fig 1). Delayed gastric emptying is defined as gastric retention of greater than 60% at 2 hours or greater than 10% at 4 hours when the standard low-fat meal is used (3,4).

Small-Bowel Scintigraphy

Scintigraphic assessment of small-bowel transit is usually performed in conjunction with gastric emptying or whole-gut transit studies. The examination involves ingestion of a solid meal labeled with ^{99m}Tc (5). Sequential imaging is performed for 6 hours, and the amount of radiotracer that reaches the terminal ileum can be quantified to measure the percentage of small-bowel emptying. Studies have shown normal small-bowel transit values to be more than 50% at 6 hours (6). Delayed small-bowel transit occurs when radiotracer persists in loops of small bowel at 6 hours, with little to no activity in the terminal ileum (Fig 2).

Colon Scintigraphy

Assessment of colonic motility uses a pH-sensitive methacrylate-coated capsule containing indium 111 charcoal particles. The coated capsule dissolves on reaching the alkaline terminal ileum and releases the radioisotope into the colonic lumen (7). Anterior and posterior images are obtained with a camera at 4, 24, and 48 hours after ingestion to evaluate colonic transit.

The colon is divided into anatomic regions: segment 1 = ascending colon, segment 2 = transverse colon, segment 3 = descending colon, and segment 4 = rectosigmoid. Segment 5 refers to the expelled radioactive tracer. The geometric center is used to measure progression of radiotracer activity and represents a weighted average of the counts in each region (Fig 3).

For example, a low geometric center indicates that the mean activity is in the proximal colon, while a higher geometric center indicates that radiotracer has passed to the left side of the colon or been eliminated in stool. Normal values for geometric center have been established as 1.7–4.0 at 24 hours and 3.0–4.8 at 48 hours (8). Retention of tracer throughout the entire colon suggests slow-transit constipation, whereas concentration of tracer in the rectosigmoid colon could suggest an evacuation disorder.

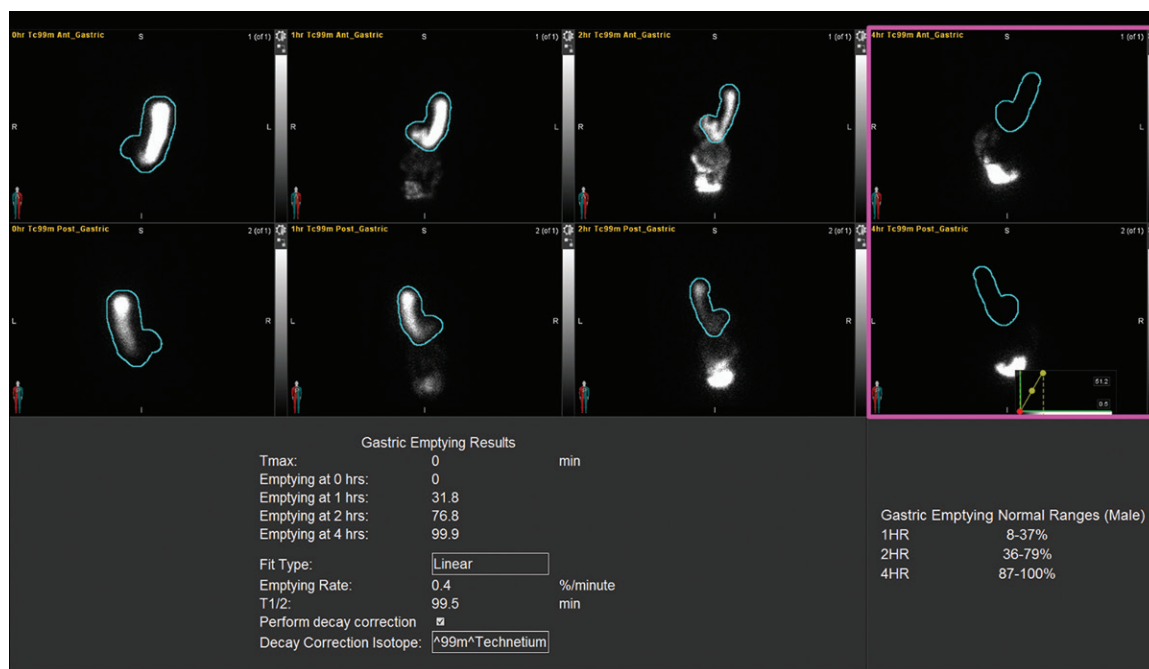
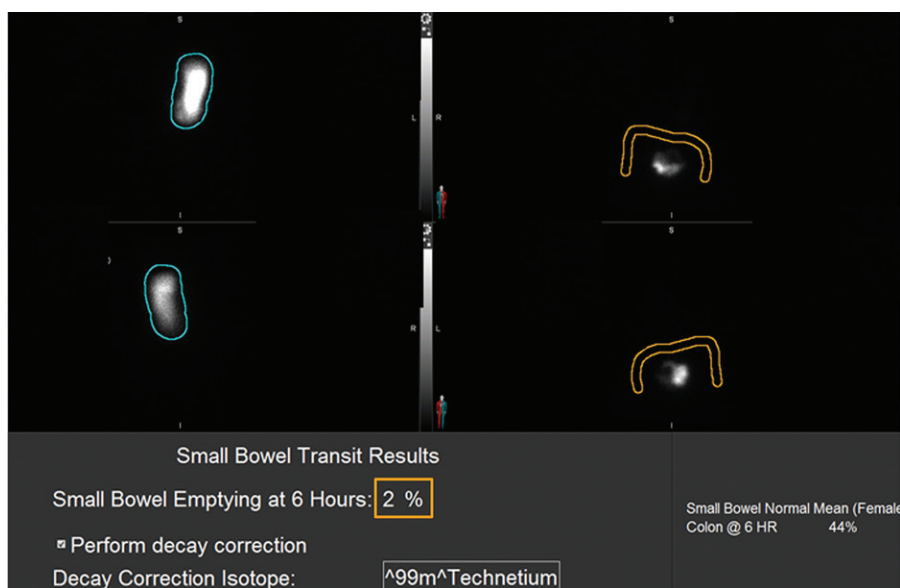


Figure 1. Normal results of gastric-emptying scintigraphy in a 34-year-old man with diabetes. Imaging is performed in the anterior and posterior projections immediately after meal ingestion and at 1, 2, and 4 hours. Imaging at 4 hours shows normal emptying of radiotracer from the stomach (pink box). Normal gastric emptying at our institution is defined as emptying of 87%–100% of the contents after 4 hours.

Figure 2. Abnormal results of small-bowel scintigraphy in a 46-year-old woman with abdominal pain and bloating. Small-bowel emptying at 6 hours after meal ingestion is 2% (orange box), thereby indicating abnormal small-bowel transit. Left: Images show radiotracer within the small bowel (blue outline) in the anterior and posterior projections at initial ingestion. Right: Images show minimal passage of radiotracer into the colon (orange outline) after 6 hours.



Gastric Accommodation Testing

Assessment of gastric accommodation is performed with SPECT. After intravenous administration of ^{99m}Tc-sodium pertechnetate, which is taken up by gastric mucosa, three-dimensional SPECT volumetric imaging of the stomach is performed 10 minutes after injection for assessment of fasting volume. Subsequently, the patient consumes 300 mL of a nutrition drink (Ensure; Abbott Laboratories), followed by a second postprandial SPECT acquisition. The change in gastric volume between the post-

prandial and fasting states represents gastric accommodation.

Esophagography

Fluoroscopic evaluation of the esophagus allows both morphologic assessment of the esophagus as well as qualitative assessment of motility (9). Esophagography may be initially used in evaluation of patients with dysphagia to exclude a structural abnormality. During barium esophagography, the radiologist evaluates the esophageal mucosa and esophageal emptying with the

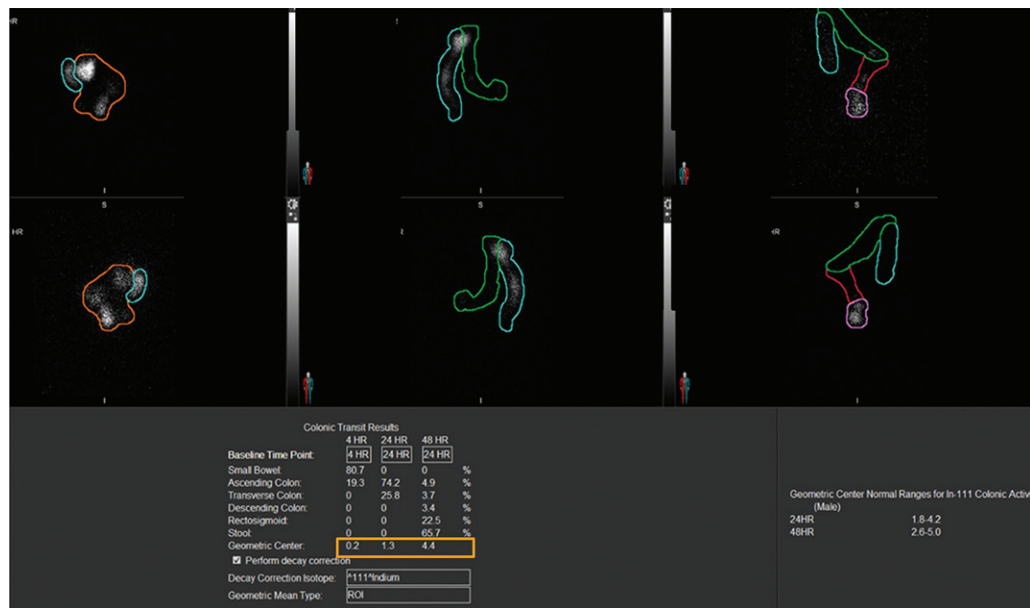


Figure 3. Normal results of colon scintigraphy in a 36-year-old man with constipation. Images are obtained at 4 hours (left), 24 hours (middle), and 48 hours (right) in the anterior (top) and posterior (bottom) projections. Results after 48 hours show a geometric mean of 4.4 (orange box), indicating that the majority of radiotracer has passed to the rectosigmoid colon.

patient in the upright position. Subsequently, with the patient in the recumbent right anterior oblique (RAO) position, the radiologist observes single swallows to evaluate esophageal motility and rapid sequential swallows to assess for strictures or rings. Gastroesophageal reflux may also be identified with the patient in the recumbent position.

Timed barium esophagography is performed to assess esophageal emptying in cases of suspected achalasia or for objectively assessing treatment response. The patient swallows a fixed volume of barium suspension while in the upright position, and left posterior oblique images are typically obtained at 1, 2, and 5 minutes after ingestion to measure the height of the retained barium column and the rate of emptying. Improvement in esophageal emptying after treatment of achalasia can be accurately evaluated by using the same protocol for follow-up esophagography (Fig 4).

Upper Gastrointestinal Series

An upper gastrointestinal series uses barium to evaluate the stomach and duodenum and is used for evaluation of abdominal pain, early satiety, dyspepsia, or postoperative anatomy. The examination is performed with the patient in multiple recumbent obliquities to allow detection of morphologic abnormalities of the gastric cardia, fundus, antrum, and pylorus and the duodenum. Upper gastrointestinal series are useful for assessing mechanical obstruction and can suggest gastroparesis by demonstrating lack of peristalsis, poor emptying at qualitative assessment, and a

distended stomach with retained debris despite adequate fasting (10).

Small-Bowel Follow-through Study

A small-bowel follow-through study allows exclusion of mechanical obstruction, strictures, and diverticula, which can all cause delayed small-bowel transit. Fluoroscopic evaluation allows qualitative assessment of intestinal peristalsis and transit time from the duodenum to the cecum. The limitation of this technique is related to lack of data regarding the normal transit time of barium in the small bowel, although the largest study reported that 83% of patients had a transit time of less than 2 hours (11).

Radiopaque Marker Study

Radiopaque markers may be used to estimate colonic transit in patients with chronic constipation. The examination involves ingestion of a gelatin capsule containing radiopaque rings, followed by abdominal radiography to count the number of markers remaining in the abdomen. One option is to perform abdominal radiography 5 days after ingestion of a capsule with 24 markers. Normal transit is characterized by five markers (<20%) or fewer retained after 5 days (Fig 5) (12).

Defecography

Defecography allows anatomic and functional evaluation of the rectum and anus. Oral ingestion of barium approximately 45–60 minutes before the study helps in identification of an

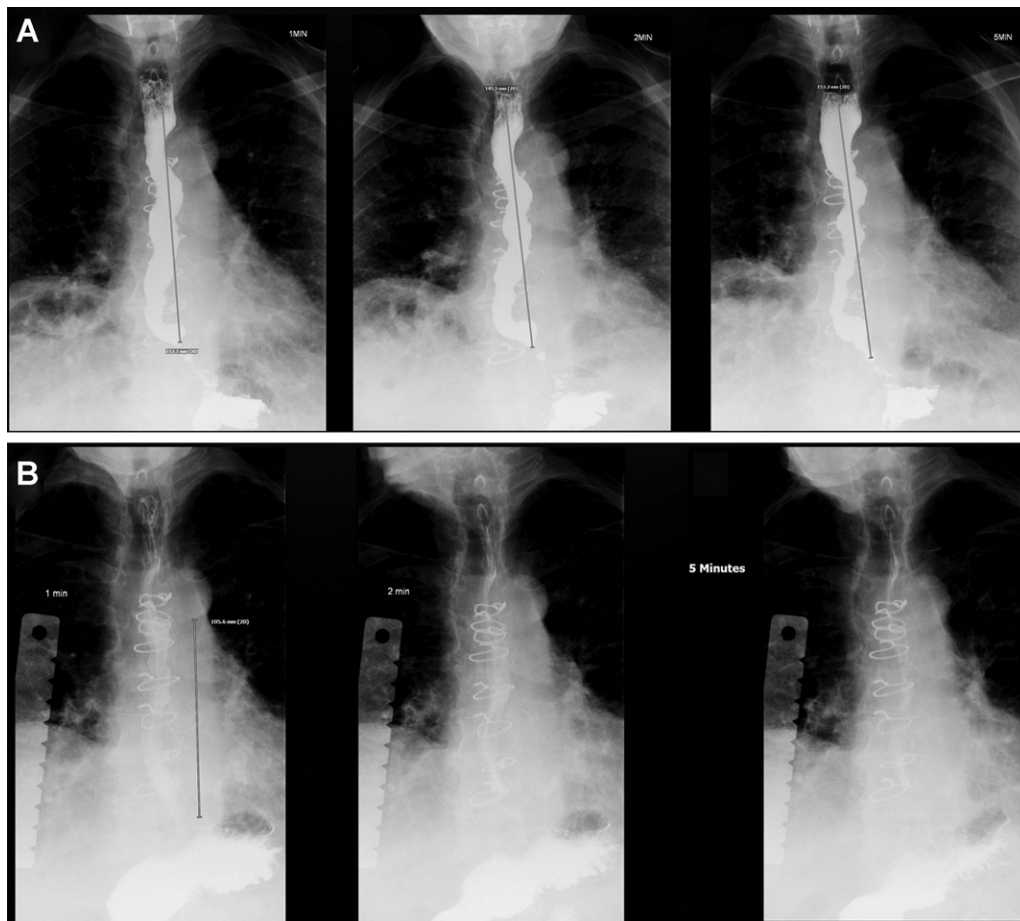


Figure 4. Timed barium esophagography in a 76-year-old man with a history of type III achalasia. **(A)** Timed esophagrams at 1, 2, and 5 minutes after ingestion of barium show a standing column of barium measuring 14–15 cm and smooth narrowing at the gastroesophageal junction. Subsequently, the patient underwent peroral endoscopic myotomy for treatment of achalasia. **(B)** Follow-up timed barium esophagrams show no residual barium column at 2 and 5 minutes after barium ingestion.

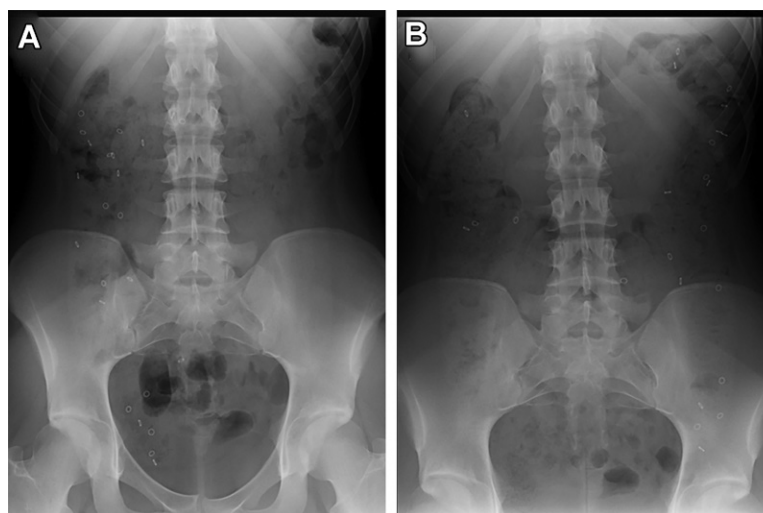


Figure 5. Radiopaque marker study in a 20-year-old woman with constipation. **(A)** Initial radiograph at day 1 shows markers projecting over the ascending and proximal transverse colon. **(B)** Subsequent radiograph obtained after 5 days shows retained markers projecting over the distal transverse and descending colon.

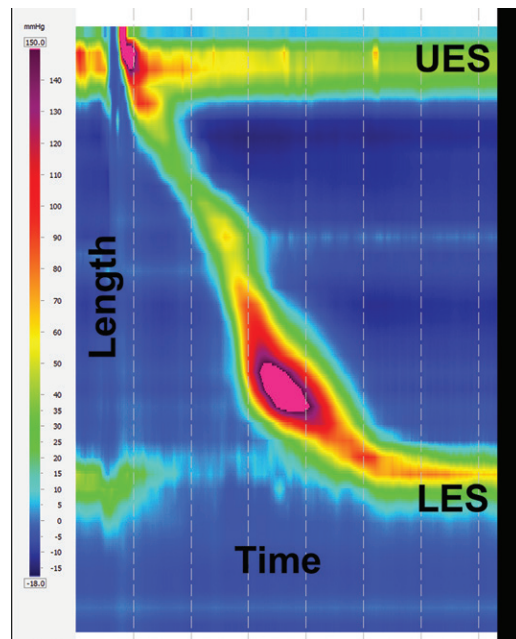


Figure 6. Normal results of esophageal manometry in a 44-year-old man. Image from esophageal manometry shows a single swallow with normal esophageal body peristalsis and normal lower esophageal sphincter (LES) relaxation. After calibration and application of a topical anesthetic, the high-resolution esophageal manometry catheter is placed transnasally. Analysis is based on 10 supine liquid swallows (5 mL of water) to evaluate both the upper esophageal sphincter (UES) and LES, intraluminal esophageal pressure, and esophageal peristalsis. Pressure is assessed in relation to time (horizontal axis) and length from the pharynx to the LES (vertical axis). Pressure is displayed as a heat map, with red representing higher pressures and blue representing lower pressures. (Courtesy of Don [Chamil] C. Coddipilly, MD, Mayo Clinic, Rochester, Minn.)

enterocele. With use of fluoroscopy, about 120–180 mL of thick barium paste is injected into the rectum with the patient in the left lateral decubitus position. Subsequently, with the patient sitting on a commode, fluoroscopic clips or spot images are obtained at rest and during pelvic floor contraction (squeezing), performance of the Valsalva maneuver, and expulsion of contrast material.

The anorectal angle (normally 108° – 127° at rest), the location of the anorectal junction, rectal evacuation, and structural abnormalities (eg, rectocele and enterocele) are observed during the examination. Pelvic floor contraction is associated with contraction of the puborectalis muscle, elevation and anterior motion of the anorectal junction, and narrowing of the anorectal angle. Conversely, defecation is associated with relaxation of the puborectalis muscle and anal sphincter, a more obtuse anorectal angle, opening of the anal canal, and rectal emptying (13).

Dynamic MR defecography is also used to assess functional disorders of the pelvic floor, which may manifest as constipation, incontinence, or prolapse. The patient can be imaged supine in a closed-magnet system or sitting in an open-magnet system. Sonographic gel is introduced into the rectum, and dynamic imaging is performed by using a steady-state sequence in the midsagittal plane with the patient at rest and during squeezing and evacuation.

The anorectal angle is measured as an indicator of relaxation of the puborectalis muscle. A key advantage of MRI is the ability to visualize the entire pelvic floor to assess for weakness

of the anterior, middle, and posterior compartments. In many cases, multicompartiment abnormalities exist together and provide useful information for both gastroenterologists and surgeons (14).

Cine MR Enterography

Cine imaging in MR enterography is used to provide functional information about small-bowel motility (Movie 1). When small-bowel motility in healthy volunteers is compared with that in patients with chronic intestinal pseudo-obstruction (CIPO), cine images demonstrate increased luminal diameter and impaired contractility in patients with CIPO (15).

Computed Tomography

CT is most helpful in identification of structural or organic causes of dysmotility. Common findings may include diffuse dilatation of the small bowel and colon or the small-bowel feces sign (particulate matter intermixed with gas bubbles in the lumen of dilated small intestine), which is suggestive of obstruction (16).

Manometry

High-resolution esophageal manometry is the standard of reference for diagnosing primary esophageal motility disorders and uses closely spaced pressure-sensitive sensors (17). The catheter extends from the hypopharynx to the proximal stomach and simultaneously measures pressure along the entire length of the esophagus. Measurement of amplitude and propagation of pressure waves is used to demonstrate the peristaltic sequence (Fig 6).

The functional lumen imaging probe (FLIP) is a newer device that uses balloon distention to assess the cross-sectional area of the esophagus during volume-controlled distention. The FLIP has been useful in cases where findings at manometry and esophagography are discordant for achalasia and to assess for esophagogastric junction outflow obstruction (EGJOO) owing to limited distensibility at the gastroesophageal junction. However, its role in clinical evaluation of esophageal motility is evolving (18).

Small-Bowel Manometry

Catheter-based manometry allows assessment of small-bowel contraction patterns and differentiation of the pathophysiology of underlying dysmotility. Myopathic disorders, including systemic sclerosis and visceral myopathy, are characterized by low-amplitude but normally propagated intestinal contractions within the affected segment (19). In contrast, neuropathic disorders such as autonomic neuropathy demonstrate normal-amplitude but uncoordinated (abnormally propagated) contractions (20,21).

Anorectal Manometry

Indications for anorectal manometry include diagnosis of dyssynergic defecation in constipated patients and identification of reduced anal sphincter function in patients with fecal incontinence (22).

Esophagus

Esophageal peristalsis is mediated by several different mechanisms. The proximal one-third of the esophagus is composed of striated muscle, the distal one-third is composed of smooth muscle, and the middle one-third contains a mixture of striated and smooth muscle. Peristalsis in the striated muscle is controlled by the lower motor neurons in the nucleus ambiguus of the brainstem. In contrast, local excitatory and inhibitory neuronal activation controls peristalsis of the smooth muscle segments (23). The lower esophageal sphincter (LES) consists of smooth muscle, which prevents gastroesophageal reflux when contracted and allows passage of food into the stomach when relaxed.

Esophageal dysmotility may manifest as nonspecific symptoms, including dysphagia, chest pain, and regurgitation (Tables 1, 2). The Chicago Classification version 4.0 of esophageal motility is an algorithmic approach to diagnosis of motility disorders (Table 3) (24).

Achalasia

Achalasia is characterized by failure of the LES to relax and absent peristalsis in the smooth

muscle portion of the esophagus due to degeneration of ganglion cells in the myenteric plexus (25). The etiology of primary achalasia is unknown. Secondary achalasia (pseudoachalasia) may be caused by infiltrative malignancy at the gastroesophageal junction, circulating lung cancer-related antibodies, or Chagas disease.

Compared to patients with primary achalasia, patients with secondary achalasia are older at presentation, report a shorter duration of symptoms, and often have significant short-term weight loss. Additionally, the narrowed LES segment is longer with little or no proximal esophageal dilatation (26). Conditions that mimic achalasia include (a) distal strictures that have induced aperistalsis and (b) scleroderma associated with reflux-induced strictures. In both cases the myenteric plexus is intact, but the LES is strictured and increasing luminal pressure is ineffective at opening it.

The most frequent symptoms of primary and secondary achalasia include progressive dysphagia to solids and liquids, heartburn, regurgitation, and noncardiac chest pain. The diagnosis should be considered in patients presenting with dysphagia, heartburn that is unresponsive to proton-pump inhibitor therapy, retained food in the esophagus at endoscopy, and resistance to endoscope passage across the gastroesophageal junction. In addition, patients with achalasia are at increased absolute risk for development of esophageal squamous cell carcinoma (27).

Barium esophagography typically demonstrates a bird-beak deformity of the distal esophagus, caused by the persistently contracted LES and aperistalsis in the mid and distal esophagus, often with tertiary contractions and an air-fluid level (Fig 7). Without treatment, patients can progress to end-stage achalasia, in which the esophagus appears markedly dilated and tortuous with a sigmoid shape (Fig 8). Pseudoachalasia demonstrates similar characteristics at barium esophagography; therefore, it is important to evaluate the gastric cardia for a potential infiltrating mass (Fig 9).

Additionally, epiphrenic diverticula in the distal esophagus are frequently associated with motility disorders and are commonly seen in association with achalasia owing to increased intraluminal pressure (Fig 10) (28).

Manometry allows confirmation of the diagnosis of achalasia by showing characteristic pressure changes. The three types of achalasia are all characterized by absence of normal peristalsis but vary in regard to esophageal contractility and patterns of pressurization (Fig 11). Type I achalasia shows lack of esophageal peristalsis and a quiescent esophageal body. Type II achalasia (the most com-

Table 1: Clinical and Imaging Features of Esophageal Motility Disorders

Condition	Clinical Features	Imaging Findings	Manometric Findings
Achalasia	Dysphagia to solids and liquids, heartburn, regurgitation, noncardiac chest pain	“Bird-beak” deformity of distal esophagus caused by persistently contracted LES; absence of peristalsis in mid and distal esophagus	Type I: absence of esophageal peristalsis Type II: periods of simultaneous panesophageal pressurization due to contraction of muscles of distal esophagus in 20% of swallows Type III: premature or spastic distal esophageal contractions in 20% of swallows
EGJOO	Dysphagia, noncardiac chest pain	Barium esophagography may help identify a structural cause (eg, stricture or malignancy) as cause of symptoms	Elevated median integrated relaxation pressure (IRP),* peristalsis not meeting criteria for achalasia
Scleroderma	Gastroesophageal reflux, dysphagia	Impaired or absent peristalsis with dilatation of distal esophagus, gastroesophageal reflux, rapid flow of contrast material in upright position due to a patulous LES	Hypotensive LES with low resting sphincter pressure and either low-amplitude contractions or aperistalsis
Distal esophageal spasm (DES)	Dysphagia to solids and liquids, noncardiac chest pain; symptoms are intermittent	Severe nonperistaltic contractions, resulting in corkscrew or rosary bead appearance of distal esophagus	Simultaneous premature esophageal contractions in 20% of swallows, with normal LES relaxation
Hypercontractile esophagus	Dysphagia, chest pain	No characteristic imaging findings	Distal contractile vigor [†] of >8000 mm Hg · cm · sec
Post-obesity surgery esophageal dysfunction (POSED)	Dysphagia	Absence of esophageal peristalsis	Increased gastric pressure, normal relaxation at gastroesophageal junction

* Integrated relaxation pressure (IRP): Measures relaxation pressure across the esophagogastric junction. Defined as the mean of the 4-second maximal relaxation in the 10-second deglutitive window beginning at UES relaxation.

[†] Distal contractile vigor: Measures contractile vigor, which is the product of contraction amplitude × duration × length from the transition zone to the proximal margin of the esophagogastric junction. Normal range = 450–8000 mm Hg · cm · sec.

mon type) demonstrates periods of simultaneous panesophageal pressurization due to contraction of the muscles of the distal esophagus. Type III achalasia demonstrates premature or spastic distal esophageal contractions with or without periods of panesophageal pressurization.

The treatment options for achalasia are primarily aimed at reducing the LES resting pressure. Preferred options include pneumatic dilation and myotomy (laparoscopic or endoscopic) in patients without contraindications to surgery. In patients who are poor surgical candidates, injection of botulinum toxin lowers LES pressure, although the effect is temporary. Patients with type III achalasia exhibit obstructive contractility of the distal esophagus and are less responsive to pneumatic dilation and myotomy than patients with type I or II

achalasia. Peroral endoscopic myotomy (POEM) is the preferred treatment in type III achalasia to tailor the length of the myotomy to include the involved smooth muscle (29).

Esophagogastric Junction Outflow Obstruction

Primary and secondary EGJOO are disorders caused by obstruction at the esophagogastric junction. For example, studies have demonstrated the association between long-term opioid therapy and elevated LES relaxation pressures (30). Postoperative changes related to fundoplication or compression due to hiatal hernia may also lead to EGJOO. Dysphagia is the most common presenting symptom, sometimes accompanied by noncardiac chest pain or regurgitation.

Table 2: Differential Diagnoses Based on Clinical and Imaging Features of Esophageal Motility Disorders

Differential Diagnoses	Differentiating Features	Recommended Correlative Examinations
Scenario: Dilated esophagus with decreased or absent primary peristalsis		
Scleroderma and mixed connective-tissue disease	Patulous esophagogastric junction Hiatal hernia Gastroesophageal reflux disease (GERD) Clinical history	History
Achalasia	Tapered esophagogastric junction	Manometry
Opioid-induced	Clinical history	History, response to cessation
Dysautonomia	Clinical history	History
Scenario: Decreased primary peristalsis with nonperistaltic contractions		
DES	Corkscrew appearance at esophagography Chest pain	History
Nonspecific motility disorder	Can be seen in young and older patients Asymptomatic or manifests as dysphagia Interrupted primary peristalsis Nonpropulsive tertiary contractions	History Manometry, if indicated
Type II or III achalasia	...	Manometry
Scenario: Esophagogastric junction narrowing		
Achalasia	Short tapered bird-beak appearance Absent primary peristalsis Distention with increased hydrostatic pressure	Manometry
Pseudoachalasia	Irregular contour Mass in gastric cardia Intact primary peristalsis, although with chronic obstruction motility may be absent	Endoscopy, CT
Decreased LES opening or spasm	Intermittent	Manometry, if indicated
Nonspecific motility disorder	Decreased motility with poor bolus propagation	Endoscopy

Table 3: Chicago Classification of Motility Disorders

Disorders with EGJOO
Type I achalasia
Type II achalasia
Type III achalasia
EGJOO
Disorders of peristalsis
Absent contractility
DES
Hypercontractile esophagus
Ineffective esophageal motility

The diagnosis is made with high-resolution manometry, which demonstrates an elevated median integrated relaxation pressure (IRP) at the esophagogastric junction, but with sufficient esophageal peristalsis not meeting the criteria for achalasia. In addition, the combination of obstructive symptoms and supporting evidence of obstructive physiology from timed barium esoph-

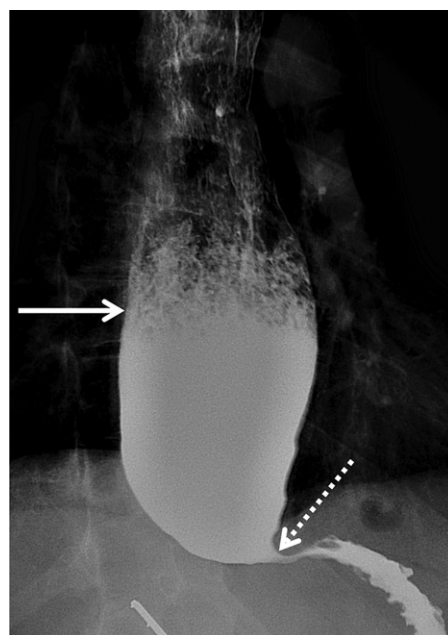


Figure 7. Achalasia in a 49-year-old man with dysphagia. Barium esophagram shows esophageal dilatation and a standing column of barium (solid arrow), with a bird-beak appearance (dotted arrow) of the gastroesophageal junction due to a persistently contracted LES.

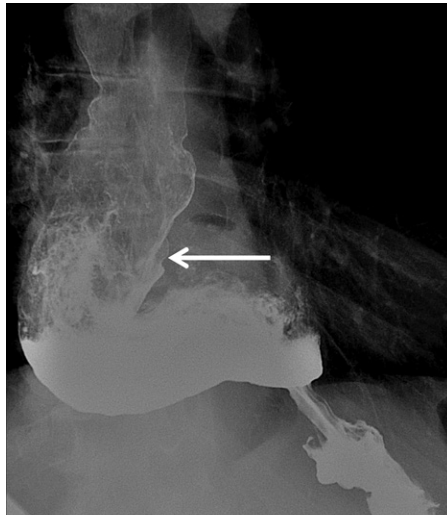


Figure 8. End-stage achalasia in a 67-year-old man with a long-standing history of dysphagia and chest pain. Barium esophagram shows a dilated esophagus (arrow) with a sigmoid-shaped appearance.

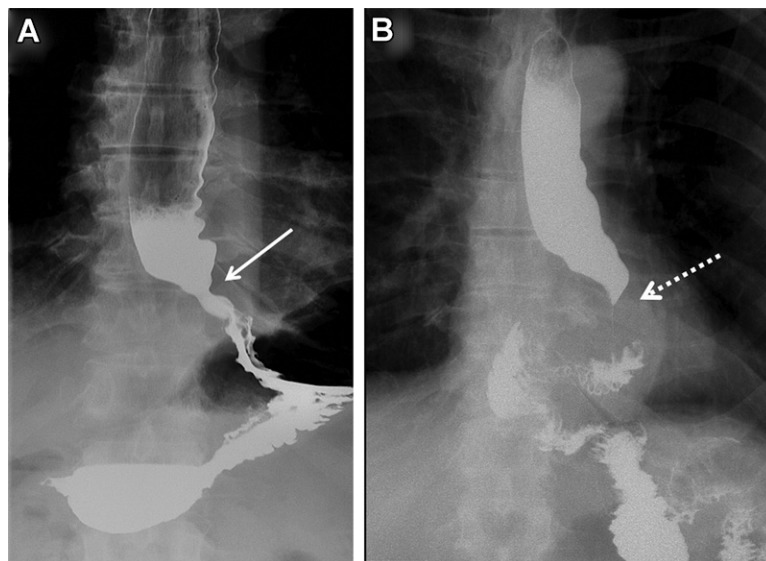


Figure 9. Pseudoachalasia in a 72-year-old man with dysphagia. (A) Esophagram obtained in 2019 shows narrowing of the distal esophagus and delayed esophageal emptying (arrow). Esophagogastroduodenoscopy performed with biopsies of the gastric cardia demonstrated poorly differentiated adenocarcinoma with signet-ring features. The patient was treated with neoadjuvant chemotherapy and subsequently underwent total gastrectomy with Roux-en-Y esophagojejunostomy. (B) Barium esophagram obtained in February 2022 shows a diffusely dilated esophagus with distal tapering at the esophagojejunal anastomosis and incomplete emptying (arrow). The findings were due to recurrent adenocarcinoma.

agography or the functional lumen imaging probe (FLIP) allows conclusive diagnosis (31,32).

Treatment of idiopathic EGJOO is based on the predominant symptom and may include conservative management if symptoms are mild or surgical myotomy if symptoms are severe.

Scleroderma

Scleroderma is a chronic multisystem disease of unknown etiology. The two subtypes include diffuse scleroderma and the CREST (calcinosis of the skin, Raynaud phenomenon, esophageal dysmotility, sclerodactyly, telangiectasia) syndrome.

Involvement of the gastrointestinal tract is seen in approximately 90% of patients with scleroderma; although the esophagus is the most frequently involved, any segment of the gastrointestinal tract may be affected. Atrophy and fibrosis of the smooth muscle in the lower two-thirds of the esophagus result in absent peristalsis and impaired function of the LES. Symptoms include

gastroesophageal reflux due to the incompetent LES and dysphagia due to dysmotility. Complications from long-standing reflux include stricture or esophagitis (33).

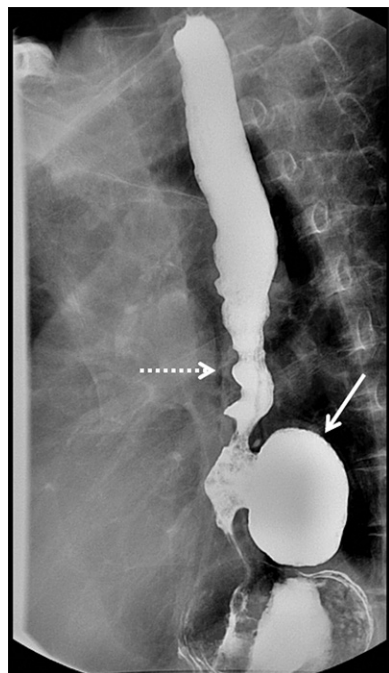
Findings at barium esophagography include impaired or absent peristalsis with associated dilatation of the distal esophagus, gastroesophageal reflux, hiatal hernia, and a patulous gastroesophageal junction (Fig 12, Movie 2). Manometric findings include a hypotensive LES with low resting sphincter pressure and either low-amplitude contractions or aperistalsis in the distal smooth muscle portion of the esophagus (Fig 13).

Treatment is aimed at improving the underlying symptoms of reflux, including use of proton-pump inhibitors.

Distal Esophageal Spasm

The pathophysiology of DES is related to impaired inhibitory innervation, leading to premature and rapidly propagated contractions of the

Figure 10. Epiphrenic diverticulum in a 78-year-old man. Esophagram shows the epiphrenic diverticulum (solid arrow). There is also severe esophageal dysmotility with absent peristalsis and nonpropulsive contractions in the distal two-thirds of the esophagus (dotted arrow).



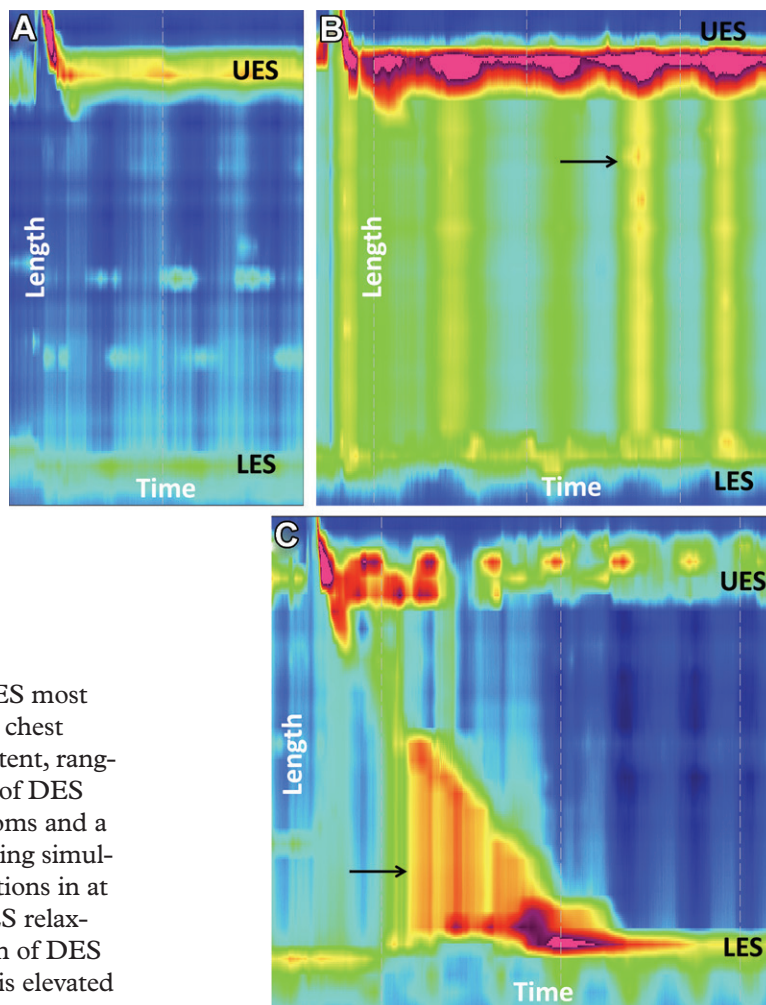
distal esophagus (34). Patients with DES most commonly present with dysphagia and chest pain; symptoms are classically intermittent, ranging from mild to severe. The diagnosis of DES requires both clinically relevant symptoms and a conclusive manometric diagnosis showing simultaneous premature esophageal contractions in at least 20% of swallows, with normal LES relaxation. This feature allows differentiation of DES from type III achalasia, in which there is elevated LES relaxation pressure.

At barium esophagography, DES may demonstrate diffuse nonperistaltic contractions, which result in a rosary bead or corkscrew appearance of the distal esophagus, although this appearance is nonspecific (Fig 14, Movie 3). Management includes symptomatic relief with pharmacologic therapy (nitrates, calcium channel blockers). Endoscopic myotomy may provide relief in patients with persistent symptoms (35).

Hypercontractile Esophagus

Hypercontractile (jackhammer) esophagus is an uncommon disorder likely caused by excessive cholinergic stimulation (36). A clinical diagnosis of hypercontractile esophagus requires both relevant symptoms (dysphagia and noncardiac chest pain) and a conclusive manometric diagnosis; esophagography is used to rule out mechanical obstruction.

Figure 11. Manometric findings for each type of achalasia, all of which demonstrate absence of peristalsis. (A) Type I achalasia in a 47-year-old man. Manometric image shows lack of esophageal peristalsis and a quiescent esophageal body. (B) Type II achalasia in a 56-year-old woman. Manometric image shows periods of simultaneous panesophageal pressurization (arrow) due to contraction of the muscles of the distal esophagus. This finding occurs in at least 20% of swallows. (C) Type III achalasia in a 43-year-old man. Manometric image shows premature or spastic distal esophageal contractions (arrow). These occur in at least 20% of swallows. (Cases courtesy of Don [Chamil] C. Codipilly, MD, Mayo Clinic, Rochester, Minn.)



tion. Manometry demonstrates high-pressure contractions in the smooth muscle of the esophagus, as demonstrated by a distal contractile vigor of greater than 8000 mm Hg · cm · sec (Fig 15) (37).

Postsurgical Dysmotility

Postoperative dysphagia can occur as a complication of bariatric surgery, known as post-obesity surgery esophageal dysfunction (POSED), or after Nissen fundoplication.

The pathophysiology of POSED is due to diminished compliance of the proximal gastric pouch in Roux-en-Y gastric bypass or high pressure created by laparoscopic sleeve gastrectomy (38). POSED demonstrates aperistalsis of the esophagus, normal relaxation at the gastro-

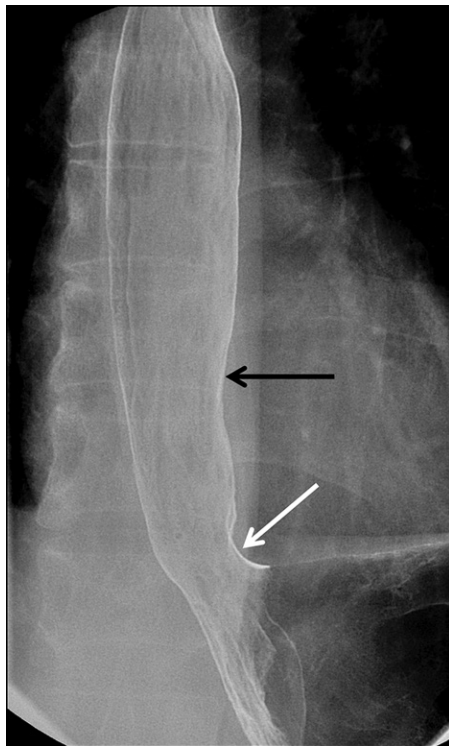


Figure 12. Scleroderma in a 42-year-old woman with gastroesophageal reflux and dysphagia. Barium esophagram with the patient upright shows patulous appearance of the gastroesophageal junction (white arrow), and moderate dilatation of the esophagus (black arrow). At fluoroscopic observation, there was absence of esophageal peristalsis (not shown).

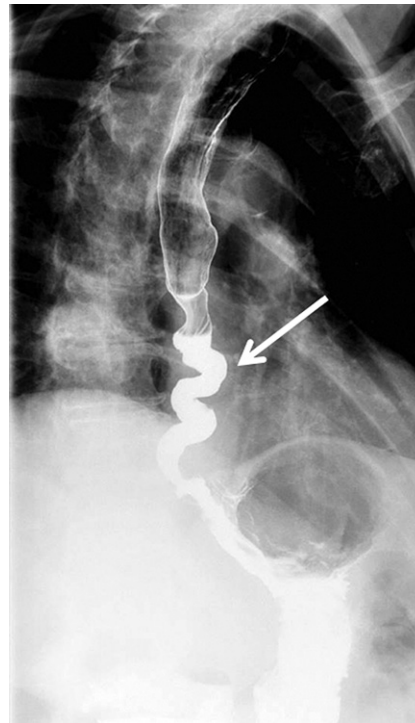


Figure 14. DES in a 53-year-old man with chest pain. Barium esophagram shows simultaneous contractions in the distal esophagus (arrow), resulting in a rosary bead or corkscrew appearance of the esophagus. The patient experienced substernal chest pain simultaneously during the examination.

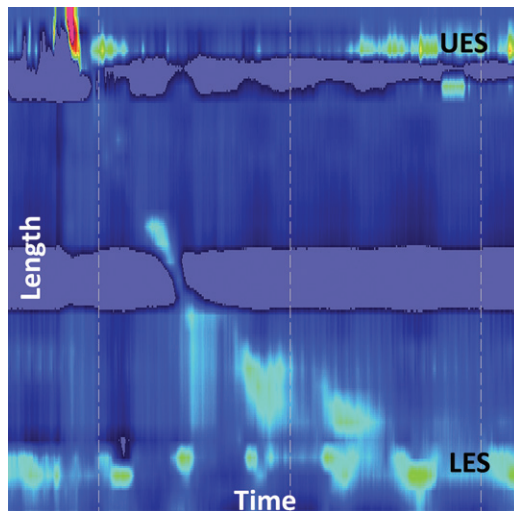


Figure 13. Scleroderma in a 59-year-old woman. Manometric image shows absent peristalsis in the esophagus and decreased LES pressure. (Courtesy of Don [Chamill] C. Codipilly, MD, Mayo Clinic, Rochester, Minn.)

esophageal junction, and increased gastric pressure (Fig 16). When the diagnosis is suspected, further investigation of the gastric pouch and anastomosis should be performed to exclude obstruction.

Dysphagia after Nissen fundoplication may be related to the obstructive effect of the fundoplication wrap. Therefore, preoperative imaging and manometry are important to assess for the presence of dysmotility (39).

Stomach

Gastric motility includes functions of the sympathetic, parasympathetic, and autonomic nervous systems. Accommodation of the gastric fundus and body is mediated by the vagus nerve, which allows relaxation so that ingested contents can be broken down by gastric acid and pepsin (40). The fundus also generates tonic and phasic contractions to transfer food to the antrum. Phasic contractions in the gastric antrum reduce solids down to small particle size, thereby allowing ingested contents to pass through the pylorus and into the small bowel as chyme. Gastric dysmotility may manifest as symptoms of postprandial nausea and vomiting and early satiety.

Gastroparesis

Gastroparesis is a syndrome of delayed gastric emptying in the absence of mechanical obstruction (Fig 17). The cause may be idiopathic, or it may be secondary to diabetes mellitus, medications (eg, narcotics), electrolyte disturbances,

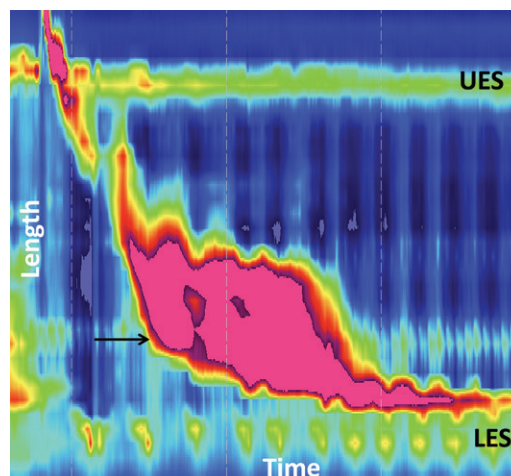
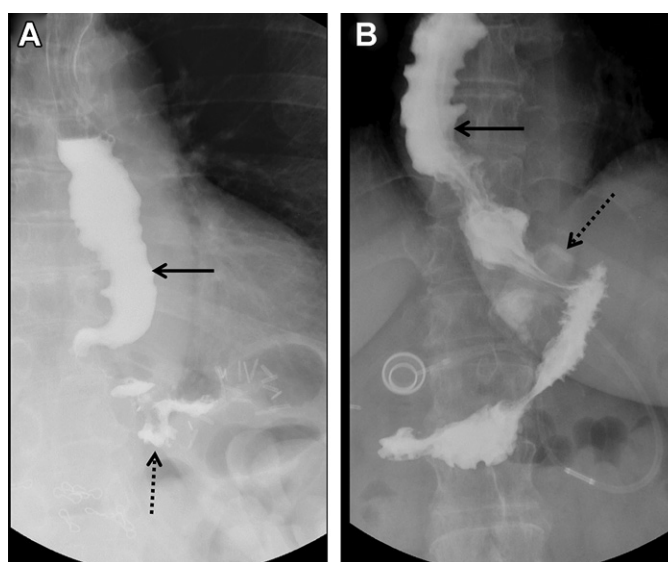


Figure 15. Hypercontractile esophagus in a 49-year-old man. Manometric image shows high esophageal pressures (arrow) but normally sequential contractions in the smooth muscle of the esophagus. (Courtesy of Don [Chamil] C. Codipilly, MD, Mayo Clinic, Rochester, Minn.)

Figure 16. Post-obesity surgery esophageal dysfunction (POSED) in two patients with dysphagia. **(A)** Barium esophagram in a 39-year-old woman shows dysmotility with repetitive nonpropulsive contractions (solid arrow) in the mid to distal thoracic esophagus after Roux-en-Y gastric bypass (dotted arrow). **(B)** Barium esophagram in a 36-year-old woman shows dysmotility with nonpropulsive contractions (solid arrow) after laparoscopic placement of a gastric band (dotted arrow).



surgery, infection, or autoimmune disorders (41). Autonomic dysfunction has also been described in patients with diabetic gastroparesis and chronic unexplained nausea and vomiting, with parasympathetic dysfunction leading to more severe upper gastrointestinal tract symptoms and delayed gastric emptying (42).

Gastroparesis is usually associated with variable upper gastrointestinal tract symptoms, including nausea, vomiting, bloating, and early satiety. The diagnosis is made with gastric-emptying scintigraphy, and gastric retention of greater than 60% at 2 hours or greater than 10% at 4 hours is considered abnormal (Fig 18). Upper gastrointestinal tract series, CT, and MRI often demonstrate a dilated and food-filled stomach without an obstructing mass.

Rumination Syndrome

Rumination syndrome is a functional gastrointestinal disorder with unknown pathogenesis. The disorder is characterized by rapid onset of effort-

less regurgitation of recently ingested food into the mouth after most meals; the regurgitation is not preceded by nausea or retching. Unlike patients with gastroparesis, patients with rumination syndrome have normal gastric emptying and accommodation.

Although no specific imaging features are present, upper gastrointestinal tract series may demonstrate considerable gastroesophageal reflux (Movie 4). Manometry can also show reflux extending to the proximal esophagus with an associated increase in gastric pressure. Treatment of rumination syndrome involves behavioral modification (eg, diaphragmatic breathing) (43).

Cannabis-induced Gastroparesis

Cannabinoids inhibit gastrointestinal motility through the CB₁ and CB₂ receptors. CB₁ receptors are located throughout the gastrointestinal tract, including within myenteric and submucosal neurons. Camilleri (44) demonstrated that dronabinol, a nonselective cannabinoid agonist,

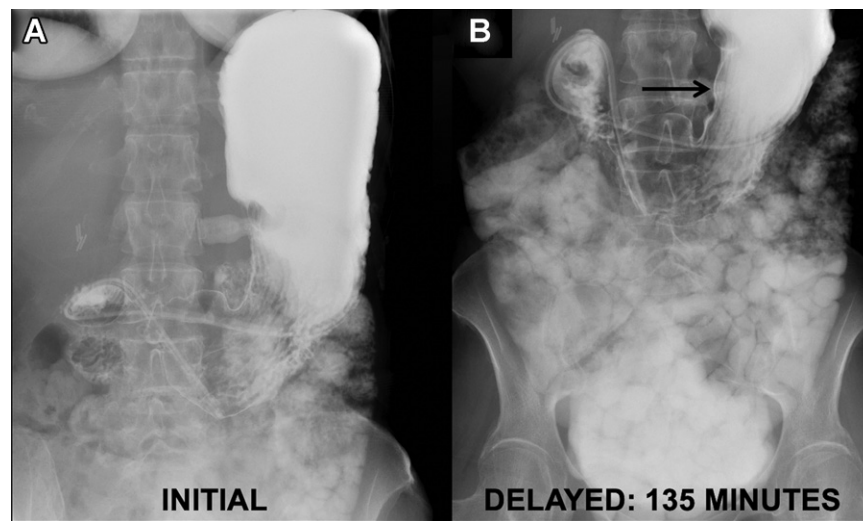


Figure 17. Gastroparesis in a 53-year-old woman with postprandial fullness and nausea. (A) Image from initial upper gastrointestinal tract series shows a large amount of barium in the stomach. (B) Delayed image at 135 minutes from a small-bowel follow-through series shows a markedly distended stomach with retained barium, without a discrete obstructing lesion at the pylorus (arrow), and passage of barium through small-bowel loops. Barium examination is helpful to exclude obstructive causes of delayed gastric emptying.

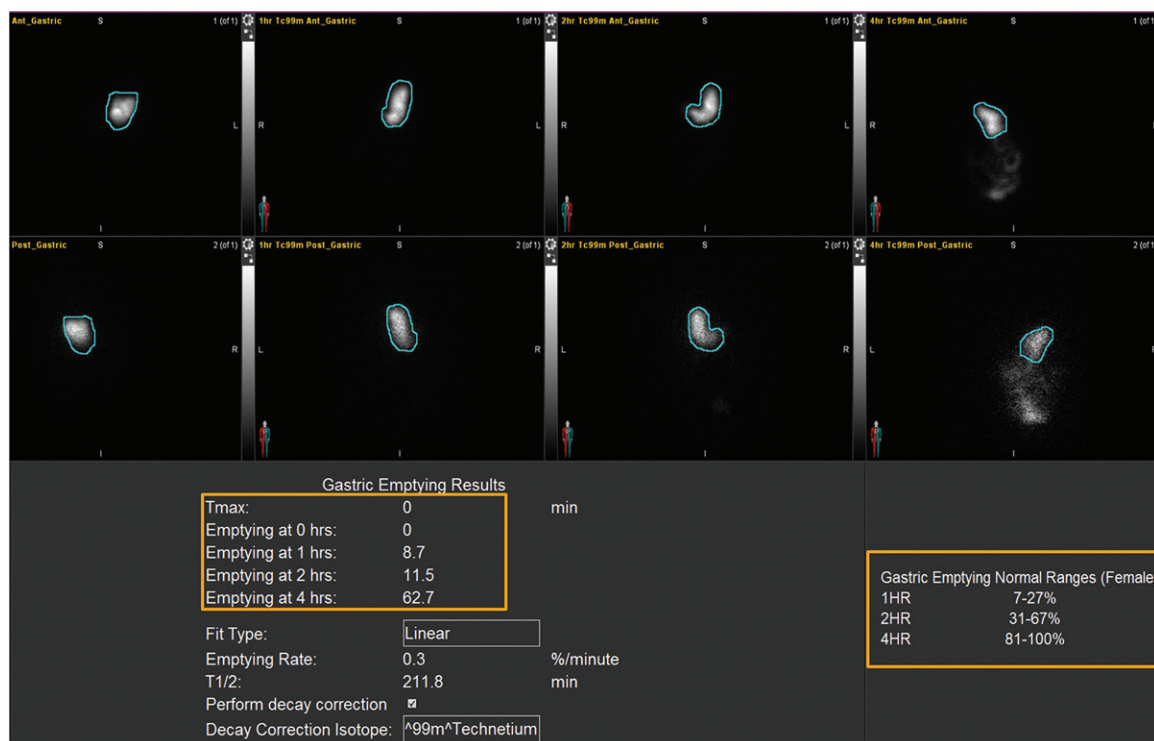


Figure 18. Abnormal results of gastric-emptying scintigraphy in a 49-year-old man with early satiety. There is delayed gastric emptying at 2 hours (11.5%) and 4 hours (62.7%) (orange boxes) after ingestion of the radiolabeled meal.

delayed gastric emptying of solids without affecting gastric accommodation.

CT may show a dilated stomach without structural obstruction (Fig 19). In addition to the effects on the stomach, cannabinoids also inhibit intestinal peristalsis, resulting in dysmotility. Long-term cannabis users may also develop cannabinoid hyperemesis syndrome, experiencing cyclic episodes of nausea and vomiting (45).

Impaired Gastric Accommodation

Patients with impaired accommodation demonstrate lack of relaxation of the gastric fundus,

leading to rapid transit from the proximal stomach to the antrum. This process causes dyspeptic symptoms, including nausea, vomiting, early satiety or postprandial discomfort, and abdominal pain (46). Impaired gastric accommodation can be associated with functional dyspepsia or diabetes mellitus and can occur after Nissen fundoplication or Billroth II gastrectomy (Fig 20).

Small Intestine

The small intestine allows digestion and absorption of ingested contents through segmentation. Interstitial cells of Cajal are responsible for the

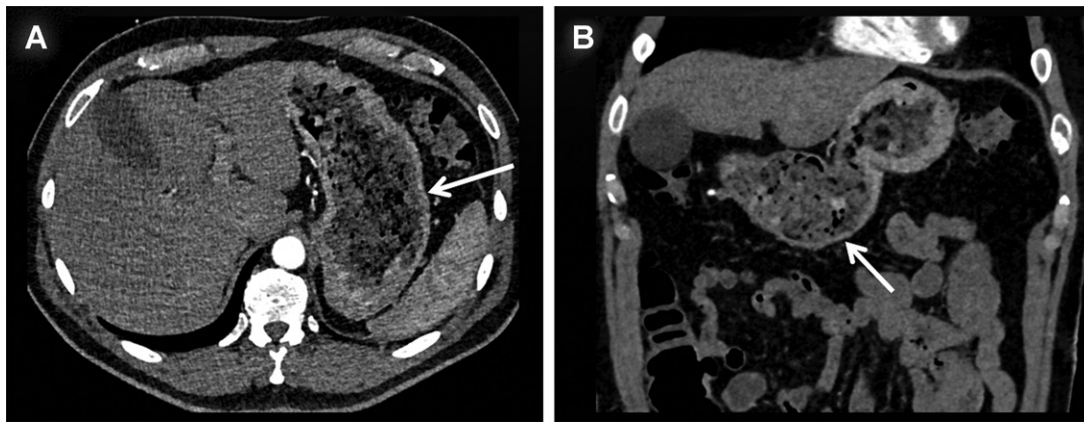


Figure 19. Cannabis-induced gastric atony in a 59-year-old man who presented to the emergency department with abdominal pain and syncope. Axial (A) and coronal (B) CT images show distention of the stomach with retained material (arrow). The patient denied any recent meal ingestion. The findings were thought to be related to long-term cannabis use.

Figure 20. Abnormal gastric accommodation at SPECT in a 42-year-old woman with early satiety and postprandial fullness. The impaired accommodation is indicated by the decreased postprandial-to-preprandial volume ratio of less than 3.0 (left orange box) and the decreased fasting to postprandial volume change of less than 428 mL (right orange box).

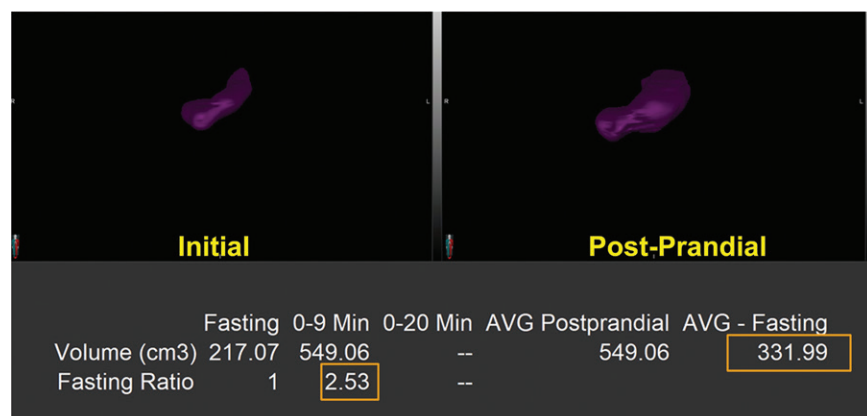


Figure 21. Intestinal pseudo-obstruction in a 55-year-old man admitted to the intensive care unit (ICU) with hypokalemia. Coronal CT image shows diffuse dilatation of the small bowel (arrows) without evidence of a transition point.



pacemaker activity of gastrointestinal motility and result in slow-wave activation of smooth muscle, thereby resulting in peristalsis (47). These phasic contractions are the basic contractile activity of the small intestine, occur more regularly in the duodenum and jejunum, and become slower and less organized in the ileum to allow absorption of nutrients (48). During the interdigestive state, the migrating motor complex (MMC) changes the motor activity of smooth muscle and causes a cyclic pattern of phasic contractions to sweep undigested residual material through the small intestine (49).

Chronic Intestinal Pseudo-obstruction

CIPO is a rare disorder characterized by dysmotility, chronic intestinal dilatation, and progressively worsening symptoms (>6 months) in the absence of mechanical obstruction (Fig

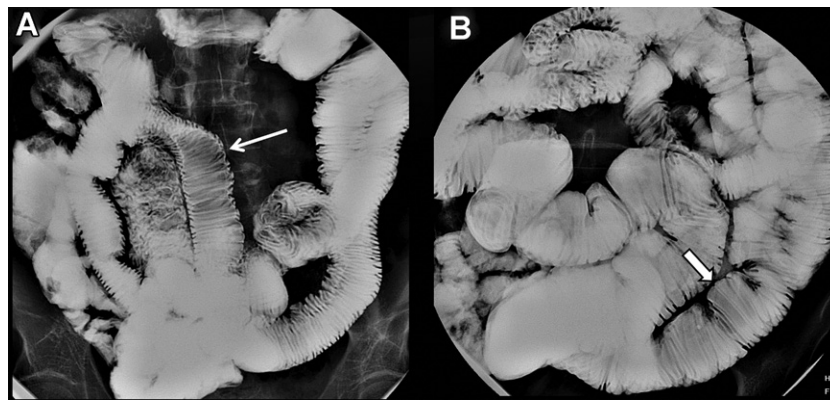


Figure 22. Scleroderma in a 42-year-old woman with abdominal pain and bloating. Images from barium small-bowel series show dilated atonic small bowel with closely spaced thin folds (arrow in A), resulting in a hidebound appearance. In addition, there are sacculations (arrow in B) along the mesenteric border of the jejunum.

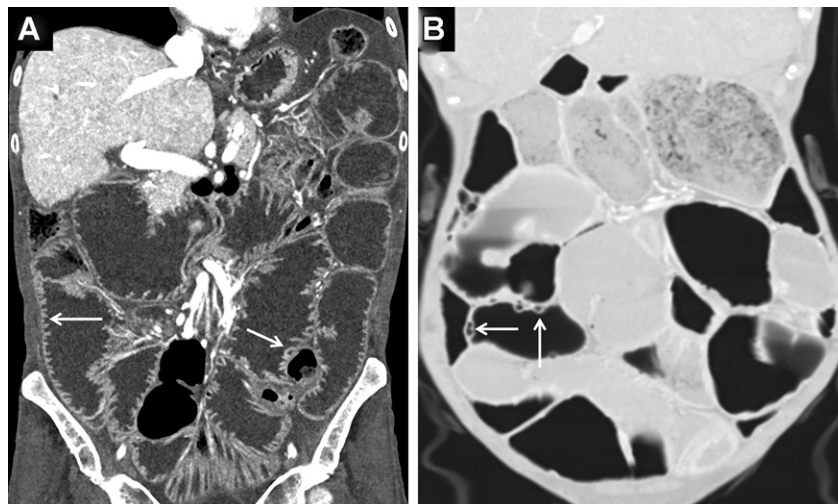


Figure 23. (A) Scleroderma in a 48-year-old woman. Coronal CT image shows marked dilatation of the duodenum and jejunum with closely spaced thin folds (arrows). The hidebound appearance results from atrophy of the inner circular muscular layer. Contraction of the outer longitudinal layer results in closely spaced valvulae conniventes. (B) Scleroderma in a 59-year-old man. Coronal CT image shows diffuse dilatation of multiple small-bowel loops with concomitant pneumatosis cystoides intestinalis (arrows), a benign finding without associated symptoms.

21); it nearly always involves the small intestine. Primary CIPO may be due to myopathy, neuropathy, or disorders affecting the interstitial cells of Cajal (mesenchymopathy). Secondary causes include systemic connective-tissue diseases (eg, scleroderma, dermatomyositis), paraneoplastic syndromes, or infections (eg, Chagas disease) that affect the autonomic nervous system or intestinal smooth muscle. Owing to the presence of nonspecific symptoms such as abdominal pain, bloating, and distention, a mechanical cause of obstruction must be excluded with radiologic imaging or endoscopy.

The most common symptoms include abdominal pain, bloating, and distention. Diagnosis is often challenging but can be achieved by using fluoroscopy or CT to rule out mechanical obstruction, laboratory studies to assess for autoimmune disorders, and assessment of gastrointestinal dysmotility and impaired transit.

Complications include small intestinal bacterial overgrowth (SIBO) and malabsorption. Treatment is focused on maintenance of nutrition, including enteral nutrition; however, patients may require gastrointestinal decompression, prokinetic agents, antibiotics, or surgery (50).

Scleroderma

Scleroderma is an acquired myopathic disorder, with the small intestine being the second most common site of gastrointestinal tract involvement after the esophagus. Dysmotility arises owing to neuropathic and myopathic changes resulting in smooth muscle atrophy and fibrosis. Studies have demonstrated reduced amplitude and frequency of the migrating motor complex (MMC), resulting in delayed small-bowel transit (51,52).

Scleroderma manifests as acute intestinal pseudo-obstruction or CIPO (luminal dilatation can be quite massive), pneumatosis cystoides intestinalis, SIBO, and asymmetric wall fibrosis resulting in wide-mouth jejunal diverticula. Patients present with recurrent or chronic abdominal pain and bloating due to hypomotility and ineffective peristalsis. In addition, SIBO leads to malabsorption, causing deficiencies of vitamin B₁₂ and fat-soluble vitamins and protein malabsorption.

Fluoroscopy and CT may show characteristic findings, including small-bowel dilatation with closely stacked valvulae conniventes (hidebound appearance of the small bowel) and sacculations often along the mesenteric border (Figs 22, 23).



Figure 24. Amyloidosis in a 43-year-old woman. Coronal CT images show generalized nodular thickening (arrows) of the small bowel, consistent with the known history of amyloidosis.

Scintigraphy is useful in detecting delayed small-bowel transit.

Amyloidosis

Amyloidosis is a rare disorder characterized by extracellular tissue deposition of abnormal fibrillar protein. The disease is usually systemic, but 10%–20% of cases may be localized within the gastrointestinal tract (53). The pathogenesis is related to mucosal and neuromuscular infiltration by amyloid protein, resulting in associated histologic and radiologic changes; there is concurrent multiple myeloma in approximately 20% of cases (54). Clinical symptoms include gastrointestinal bleeding due to the presence of mucosal lesions; malabsorption related to mucosal infiltration; chronic dysmotility with constipation, abdominal pain, bloating, or CIPO; and protein-losing gastroenteropathy manifesting as diarrhea, edema, and ascites.

CT findings of small-bowel amyloidosis are varied, including long-segment circumferential mural thickening that may result in luminal narrowing, dilatation depending on the degree of hypomotility, mesenteric infiltration with soft-tissue thickening, and lymphadenopathy (Fig 24). In addition, fluoroscopic evaluation may show nodular irregular thickening or effacement of the valvulae conniventes or loss of haustral folds (55). Diagnosis of gastrointestinal amyloidosis requires tissue biopsy. Therapy for localized disease is directed at the gastrointestinal manifestations related to dysmotility.

Enteric Visceral Myopathy

Enteric visceral myopathy is a rare genetic disorder characterized by degeneration and fibrosis of smooth muscle in the gastrointestinal tract, leading to impaired intestinal motility without findings of mechanical obstruction; it may result in CIPO (56). Involvement may be diffuse or limited to focal segments of the bowel, and patients usually

have chronic symptoms of abdominal pain, distention, nausea and vomiting, and chronic constipation (Fig 25). Diagnosis may be delayed owing to the presence of nonspecific symptoms and requires full-thickness intestinal biopsy.

Radiologic findings include esophageal aperistalsis, megaduodenum, and variable dilatation of the small and large intestine. Management of enteric visceral myopathy requires multidisciplinary therapy, including symptomatic improvement and monitoring of nutritional status.

Opioid-induced Hypomotility

Constipation is the most common and debilitating side effect of opioids. Opioids exert effects via μ receptors located in the submucosa of the small intestine (57). Stimulation of μ receptors in the enteric nervous system results in adverse gastrointestinal effects, causing nonpropulsive contractions in the small and large intestine, increased colonic fluid absorption, and stool desiccation (Fig 26).

The constipating effects of opioids are more common with long-term use and are likely dose related. The American Gastroenterological Association recommends use of laxatives as first-line therapy for opioid-induced constipation. In laxative-refractory cases, use of the μ -opioid receptor antagonists naloxegol or naldemedine is recommended over no treatment (58).

Large Intestine, Rectum, and Anus

Motility of the colon relies on phasic contractions and mass movements. Phasic contractions segment the colon into haustra, with slow mixing and retention of residue to allow absorption of water. Alternatively, mass movements are high-amplitude contractions of the circular muscular layer, which moves chyme into the rectum. Colonic peristalsis is stimulated by the gastro-

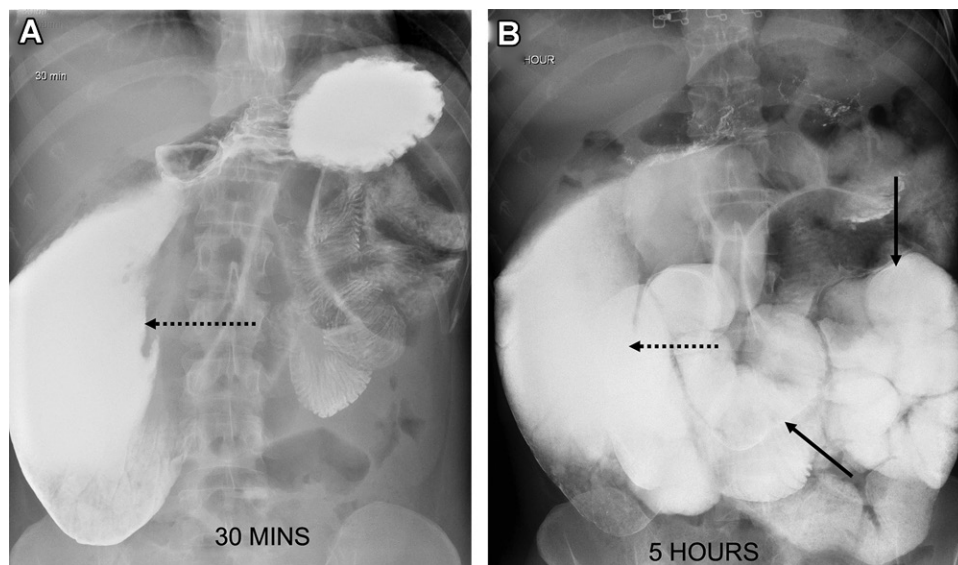


Figure 25. Enteric visceral myopathy with dilated duodenum in a 29-year-old woman referred for abdominal pain and concern for obstruction. (A) Image from small-bowel follow-through series, obtained 30 minutes after administration of barium, shows a dilated segment of duodenum (arrow) in the right abdomen. Note the narrowing of the dilated transverse duodenum by the superior mesenteric artery, a finding related to the supine positioning of the patient. (B) Delayed image from small-bowel follow-through series, obtained at 5 hours, shows the persistently dilated duodenum (dotted arrow) and dilated distal small-bowel loops (solid arrows) without passage of barium into the colon. The findings of slow transit were due to visceral myopathy.

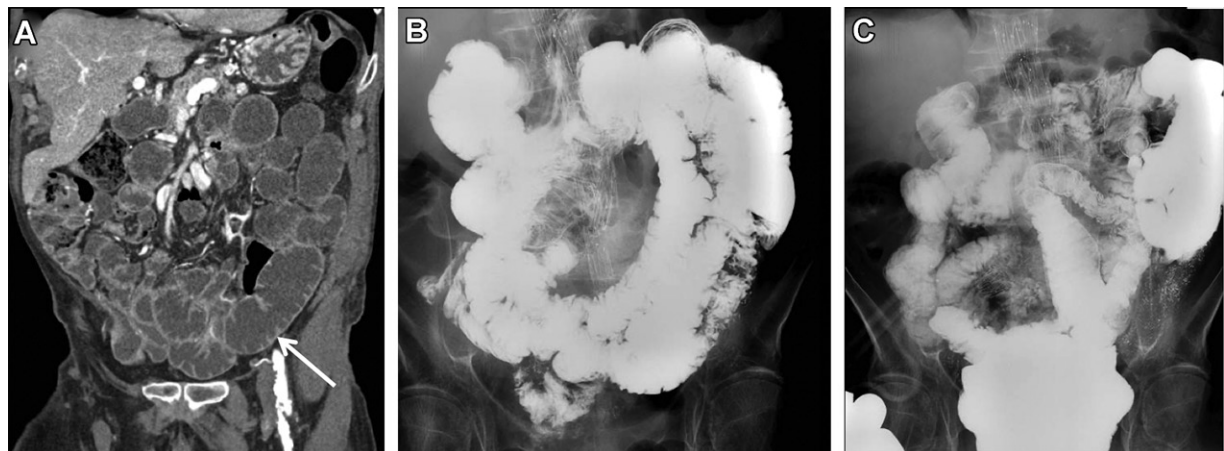


Figure 26. Opioid-induced hypomotility in a 78-year-old man with chronic abdominal pain. The patient was taking opioids for treatment of shoulder pain. (A) Image from CT enterography shows dilatation of proximal small bowel (arrow) of up to 4 cm. (B) Image from subsequent small-bowel series shows mildly dilated small bowel. (C) Image from small-bowel series, obtained at 6 hours, shows significantly delayed transit through the mid and distal ileum.

colic reflex, which occurs within 30 minutes after ingestion of a meal.

Defecation requires coordinated actions between the colon, rectum, pelvic floor, and muscles of the anal sphincter. Both the external anal sphincter and puborectalis muscle are composed of striated muscle. The internal anal sphincter is a thickened band of circular smooth muscle that is the primary determinant of the resting pressure of the anal canal. The internal anal sphincter is innervated by the autonomic nervous system and is tonically contracted at rest.

Constipation and Colonic Inertia

Constipation results in variable symptoms, including excessive straining, hard stools, infrequent bowel movements, the sensation of incomplete evacuation or anorectal obstruction, and the need for manual maneuvers to assist defecation. Chronic constipation is defined as symptoms lasting longer than 3 months. The etiology is often multifactorial.

Diagnosis relies on clinical evaluation in conjunction with additional testing, including anorectal manometry or colonic transit study.

Figure 27. Constipation in a 71-year-old man referred for abdominal distention. (A) Image from a water-soluble enema study performed with Gastrografin (diatrizoate meglumine and diatrizoate sodium solution; Bracco Diagnostics) shows a distended colon filled with stool (arrows). (B) Postevacuation image shows a large amount of retained stool (arrows) throughout the colon.

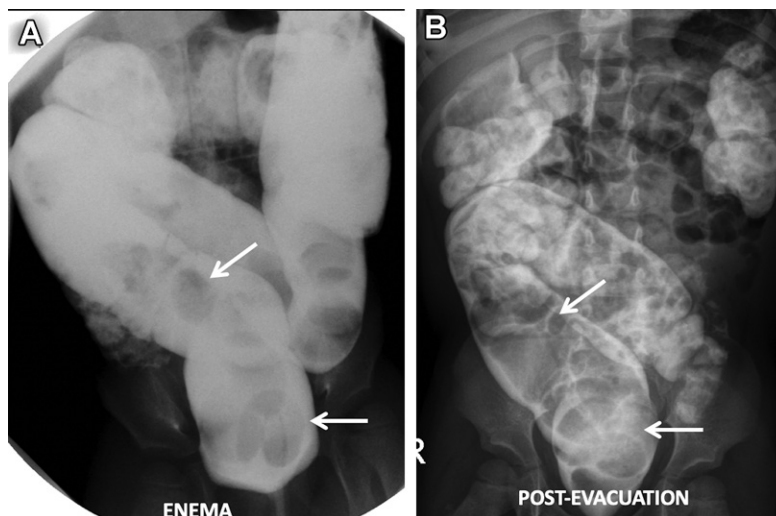
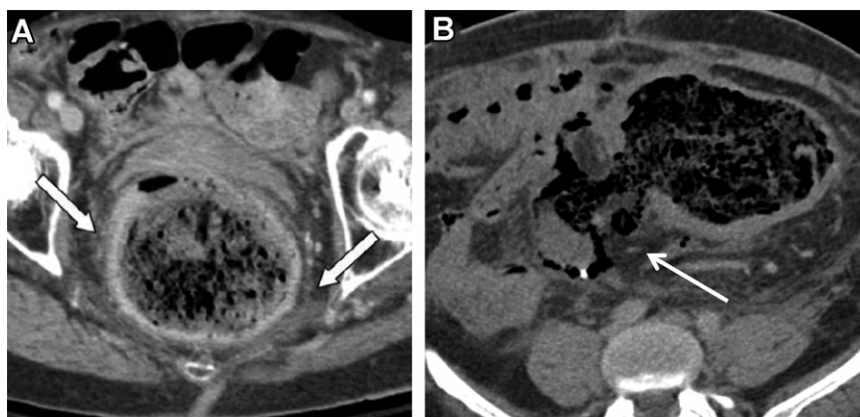


Figure 28. (A) Stercoral proctitis in a 68-year-old man with chronic constipation. Axial CT image shows rectal distention, concentric wall thickening, mural hyperenhancement, and stranding of perirectal fat (arrows). (B) Perforated stercoral colitis in a 77-year-old man with chronic constipation and new-onset abdominal pain. Axial CT image shows a defect in the wall of the colon with extraluminal gas (arrow).



However, further workup with colonoscopy must be performed in patients with unexplained weight loss, hematochezia, acute onset of constipation, or a family history of colorectal cancer. Fluoroscopic contrast enema examination is used to assess for anatomic abnormalities but does not allow assessment of the underlying cause of constipation (Fig 27). Initial management of constipation includes dietary modification and use of laxatives.

Colonic inertia may also lead to stercoral colitis, an inflammatory process of the colonic wall—usually the rectosigmoid colon—caused by increased intraluminal pressure from impacted fecal material. CT findings include colonic distention with concentric wall thickening, mural hyperenhancement, and pericolic or rectal stranding (Fig 28). Stercoral colitis may lead to localized ischemic changes, ulceration of the rectosigmoid mucosa, and ultimately colonic perforation, which has a reported mortality rate as high as 60% (59).

Acute Colonic Pseudo-obstruction

Acute colonic pseudo-obstruction (Ogilvie syndrome) is defined as acute colonic dilatation in the

absence of mechanical obstruction. The pathogenesis is postulated to be an alteration in the autonomic nervous system affecting colonic motility, and it occurs in association with severe illness or after surgery in conjunction with metabolic abnormalities. Patients typically develop abdominal distention over 3–7 days and may also experience abdominal pain, nausea, or vomiting. Abdominal radiography and CT demonstrate colonic dilatation, with the cecum and ascending colon typically showing the most pronounced distention (Fig 29).

Water-soluble contrast enema examination or CT should be performed to exclude mechanical obstruction, and CT allows evaluation of luminal diameter and for mucosal ischemia. The osmotic agent in the water-soluble enema may help relieve the pseudo-obstruction, but this study should not be performed in patients with signs of peritonitis. The risk of colonic perforation is significant when the cecal diameter is greater than 12 cm or symptoms are present for longer than 6 days. Therefore, patients should be monitored with serial physical examinations and abdominal radiography.

Initial management includes supportive therapy. However, in patients with persistent symptoms

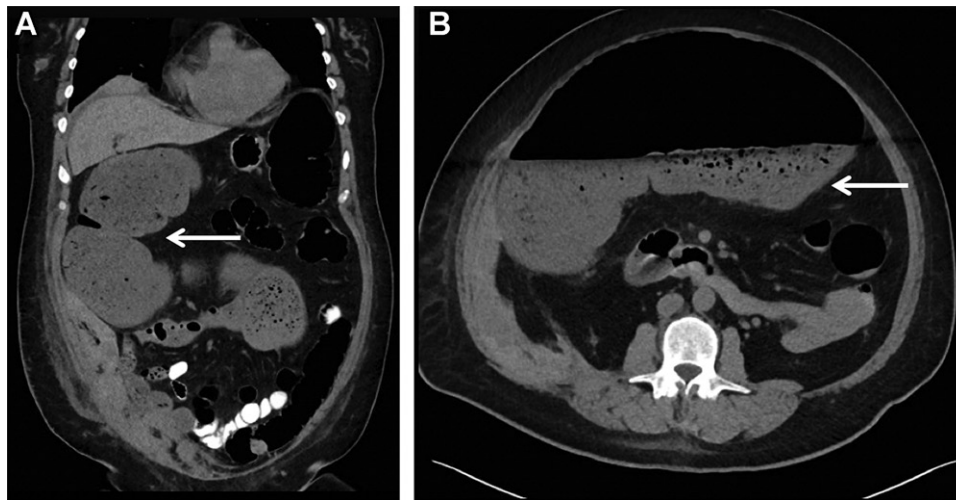


Figure 29. Acute colonic pseudo-obstruction in a 54-year-old woman with a history of end-stage renal disease who presented with nausea, bloating, and constipation. Coronal (A) and axial (B) CT images show dilatation of the entire colon to the rectum, with the anterior cecum preferentially dilated to nearly 12 cm (arrow). There is no twisting, paracolic fat stranding, or mechanical obstruction. Given the degree of cecal distention, findings were compatible with colonic pseudo-obstruction. A rectal tube was placed, and the patient was treated with neostigmine.

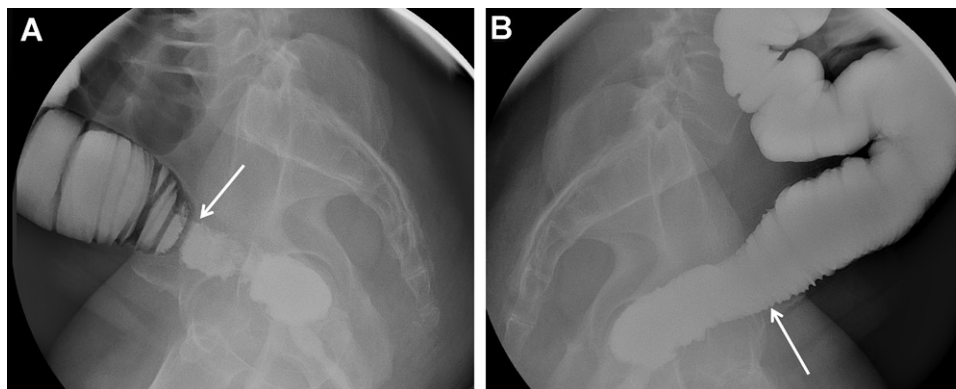


Figure 30. Hirschsprung disease in a 44-year-old man. (A) Image from water-soluble contrast enema examination shows a transition zone (arrow) between the small caliber of the distal rectum and the dilated proximal colon, with reversal of the rectosigmoid ratio. (B) Image from barium enema examination shows fasciculation or sawtooth appearance (arrow) of the mucosa of the rectum within the aganglionic segment.

and marked colonic distention, management options include intravenous administration of neostigmine or colonoscopic decompression (60).

Hirschsprung Disease

Hirschsprung disease is a congenital disorder defined by absence of the ganglion cells in the myenteric and submucosal plexuses of the distal colon due to impaired neural crest cell migration. The aganglionic segment of colon fails to relax, causing functional obstruction and upstream dilatation of the more proximal colon. Approximately 80% of cases affect the rectosigmoid colon (short-segment disease), but there may also be ultrashort-segment disease, long-segment disease, or total colonic aganglionosis. Signs and symptoms in the neonatal period include distention, constipation with failure

to pass meconium in the first 48 hours, and bilious emesis. Diagnosis in adulthood is uncommon but may be suspected in cases of recurrent constipation and abdominal distention.

Contrast enema examination typically reveals a transition zone between the normal proximal colon and the aganglionic distal segment, mucosal irregularity, and reversal of the rectosigmoid index (measurement of the diameter of the rectum divided by the diameter of the sigmoid colon, which is normally >1 in neonates) (Figs 30, 31) (61). The diagnosis is confirmed with suction biopsy, which demonstrates absence of ganglion cells and hypertrophic nerve fibers. Treatment in the majority of patients requires resection of the aganglionic segment and a pull-through procedure (62).

Figure 31. Hirschsprung disease in a 6-day-old boy. Image from water-soluble contrast enema examination shows a transition point (arrow) at the rectosigmoid junction, with a dilated proximal colon and narrowed caliber of the upper and mid rectum. The lower rectum was distended with a meconium pellet.

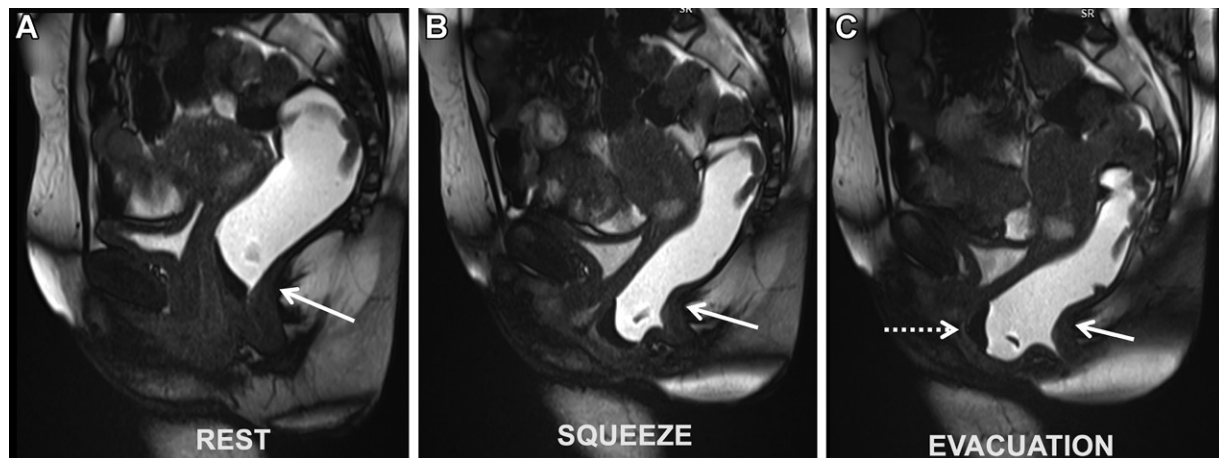
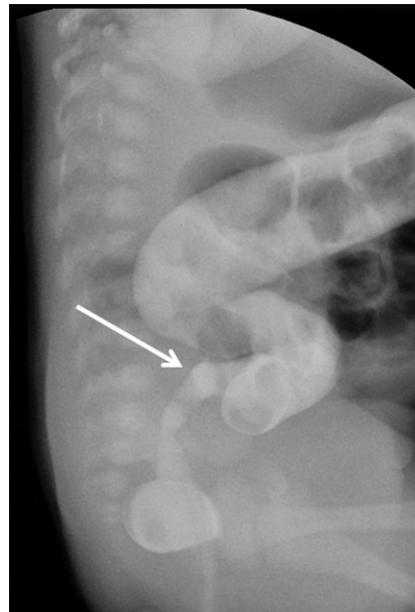


Figure 32. Results of MR defecography in a 44-year-old man with constipation. (A) Image obtained at rest shows mild impression of the puborectalis muscle (arrow). (B) Image during squeezing shows normal contraction of the puborectalis muscle (arrow). (C) Image during evacuation shows paradoxical contraction without relaxation of the puborectalis muscle (solid arrow), consistent with pelvic floor dyssynergia. In addition, a small anterior rectocele (dotted arrow) is noted.

Pelvic Floor Dysfunction

Functional obstruction of the anal canal, as seen with dyssynergic dysfunction, is an underlying cause of chronic constipation in up to 50% of patients. Dyssynergic defecation results from inappropriate contraction or lack of relaxation of the puborectalis muscle or external anal sphincter, preventing normal passage of stool. A combination of the clinical history and diagnostic tests including anorectal manometry, the rectal balloon expulsion test, electromyography (EMG), and defecography is used for diagnosis.

Delayed initiation and impaired evacuation, in addition to decreased change in the anorectal angle, are highly predictive of this diagnosis (63,64). Additionally, paradoxical sphincter contraction can be seen with impression of the puborectalis

muscle along the dorsal wall of the rectum during straining and evacuation (Figs 32, 33; Movie 5). Biofeedback therapy is the most effective treatment of dyssynergia.

Conclusion

Gastrointestinal motility disorders encompass a wide spectrum of pathologic conditions with complex pathogenesis. Symptoms are often non-specific and overlap with those of other disease processes; thus, a thorough patient evaluation is critical. The specific radiologic and gastrointestinal examinations chosen for the workup are dependent on the clinical history and symptoms. Radiologists should have an integrated understanding of the pathophysiology, as well as an understanding of both the radiologic and gastro-

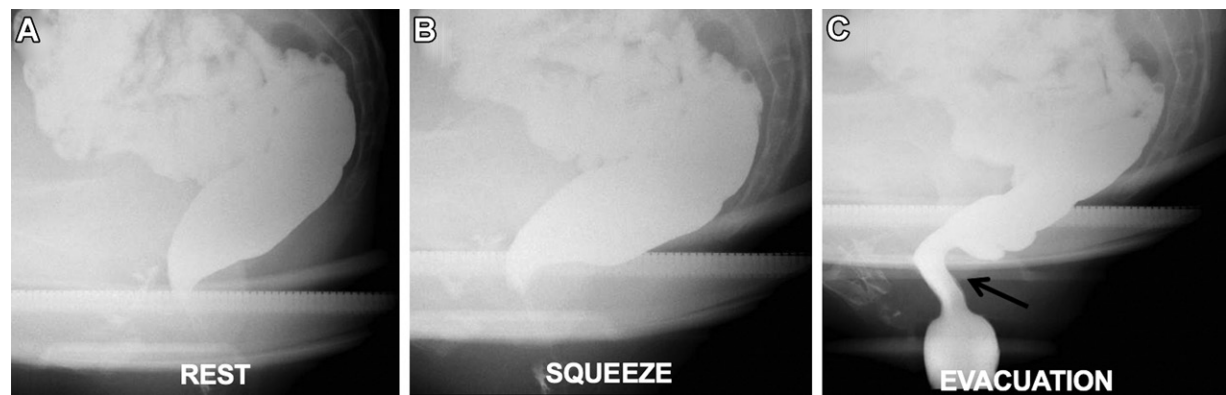


Figure 33. Results of fluoroscopic defecography for evaluation of chronic constipation in a 25-year-old woman. (A) Image obtained at rest shows normal appearance of the rectum. (B) Image during squeezing shows normal decrease of the anorectal angle. (C) Image during evacuation shows paradoxical contraction (arrow) of the puborectalis muscle. The patient was unable to completely empty the rectal contents.

enterologic findings, to effectively contribute to an accurate diagnosis.

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